MATH 231 – Linear Algebra I

The Matrix Unit, Part 5 ▪ Projection Matrices and the Direct Sum

Kiely Hall 326 ▪ Instructor: Kevin Bedoya

About this document. This is Part 5 of the matrix unit, Sections 26–29. Section numbers run continuously through the unit (Part 2's opening table lists them all). References into the vector unit always name it.

Assumed. From the vector unit: the projection $\text{proj}_{\mathbf{u}}(\mathbf{v})$ and the orthogonal piece $\mathbf{v} - \text{proj}_{\mathbf{u}}(\mathbf{v})$ (§12.4–§12.6), orthogonal vectors being independent (§21.2), and the Pythagorean identity (§23). From this unit: the inner and outer products $\mathbf{u}^{\mathsf{T}}\mathbf{v}$ and $\mathbf{u}\mathbf{v}^{\mathsf{T}}$ (§7), the laws M1–M6 (§8.4), the projection onto the x-axis (§19.3), and the reflections of §19–§21.

What you should be able to do afterwards. Write down the matrix that projects onto a given line and use it; say what idempotent means and check whether a matrix is a projection; distinguish an orthogonal projection from a slanted one; build the complementary projection and identify it in $\mathbb{R}^2$ as the projection onto the perpendicular line; build a reflection from a projection; build the Householder reflector that sends a vector onto the x-axis; and split a vector into its pieces along a direct sum.

26. The projection matrix onto a line

26.1. The formula we already have

From the vector unit. For a nonzero vector $\mathbf{u}$, the projection of $\mathbf{v}$ onto $\mathbf{u}$ is the shadow $\mathbf{v}$ casts on the line through $\mathbf{u}$ when the light falls perpendicular to that line (§12.4):

$$\text{proj}_{\mathbf{u}}(\mathbf{v}) = \frac{\mathbf{u}\cdot\mathbf{v}}{\mathbf{u}\cdot\mathbf{u}}\,\mathbf{u}.$$

The subscript names the vector projected *onto*. What is left over, $\mathbf{v} - \text{proj}_{\mathbf{u}}(\mathbf{v})$, is orthogonal to $\mathbf{u}$ (§12.6).

The formula takes a vector $\mathbf{v}$ and returns a vector. It is linear in $\mathbf{v}$: the numerator $\mathbf{u}\cdot\mathbf{v}$ is, and nothing else depends on $\mathbf{v}$. So by §13.3 there is a matrix that does it. This section finds it, not from where it sends $\mathbf{e}_1$ and $\mathbf{e}_2$, but straight from the formula.

26.2. From the formula to the matrix

Three moves, each one a law already proved.

(1) Write the dot products as matrix products, $\mathbf{u}\cdot\mathbf{v} = \mathbf{u}^{\mathsf{T}}\mathbf{v}$ (§7.1):

$$\text{proj}_{\mathbf{u}}(\mathbf{v}) = \frac{\mathbf{u}^{\mathsf{T}}\mathbf{v}}{\mathbf{u}^{\mathsf{T}}\mathbf{u}}\,\mathbf{u}.$$

(2) The fraction is a number, so it can be written on the right of $\mathbf{u}$ instead of the left (M4):

$$\text{proj}_{\mathbf{u}}(\mathbf{v}) = \mathbf{u}\,\frac{\mathbf{u}^{\mathsf{T}}\mathbf{v}}{\mathbf{u}^{\mathsf{T}}\mathbf{u}} = \frac{1}{\mathbf{u}^{\mathsf{T}}\mathbf{u}}\,\mathbf{u}\,(\mathbf{u}^{\mathsf{T}}\mathbf{v}).$$

(3) Regroup with associativity (M1), reading $\mathbf{u}$ as $n \times 1$, $\mathbf{u}^\mathsf{T}$ as $1 \times n$ and $\mathbf{v}$ as $n \times 1$: $\mathbf{u}(\mathbf{u}^\mathsf{T}\mathbf{v}) = (\mathbf{u}\mathbf{u}^\mathsf{T})\mathbf{v}$. So

$$\operatorname{proj}_{\mathbf{u}}(\mathbf{v}) = \frac{\mathbf{u}\mathbf{u}^\mathsf{T}}{\mathbf{u}^\mathsf{T}\mathbf{u}}\,\mathbf{v}.$$

The thing multiplying $\mathbf{v}$ no longer depends on $\mathbf{v}$. It is a matrix.

Definition 26.1 (Projection matrix onto a line)**.** For a nonzero $\mathbf{u} \in \mathbb{R}^n$, the **projection matrix onto the line** $\operatorname{span}\{\mathbf{u}\}$ is

$$\boxed{\mathbf{P}_{\mathbf{u}} = \frac{\mathbf{u}\mathbf{u}^\mathsf{T}}{\mathbf{u}^\mathsf{T}\mathbf{u}}} \qquad \text{so that} \qquad \mathbf{P}_{\mathbf{u}}\mathbf{v} = \operatorname{proj}_{\mathbf{u}}(\mathbf{v}) \ \text{ for every } \mathbf{v}.$$

It is the outer product $\mathbf{u}\mathbf{u}^\mathsf{T}$ (§7.2), an $n \times n$ matrix, divided by the number $\mathbf{u}^\mathsf{T}\mathbf{u} = \|\mathbf{u}\|^2$.

The matrix depends only on the *line*, not on which vector along it was chosen. Replacing $\mathbf{u}$ by $c\mathbf{u}$ multiplies the top by c^2 and the bottom by c^2. In particular, for a unit vector $\hat{\mathbf{u}}$ the denominator is 1 and $\mathbf{P}_{\mathbf{u}} = \hat{\mathbf{u}}\hat{\mathbf{u}}^\mathsf{T}$.

26.3. Examples

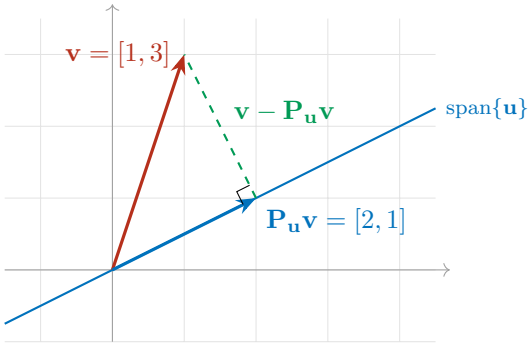
Example 26.1.

(a) $\mathbf{u} = \mathbf{e}_1 = [1, 0]$: $\mathbf{u}\mathbf{u}^\mathsf{T} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $\mathbf{u}^\mathsf{T}\mathbf{u} = 1$, so $\mathbf{P}_{\mathbf{e}_1} = \operatorname{diag}(1, 0)$, the projection onto the x-axis that §19.3 built from $\mathbf{e}_1$ and $\mathbf{e}_2$. The two constructions agree.

(b) $\mathbf{u} = [1, 1]$: $\mathbf{P}_{\mathbf{u}} = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$. For $\mathbf{v} = [3, 1]$, $\mathbf{P}_{\mathbf{u}}\mathbf{v} = \frac{1}{2}[4, 4] = [2, 2]$. The formula agrees: $\frac{\mathbf{u}\cdot\mathbf{v}}{\mathbf{u}\cdot\mathbf{u}}\mathbf{u} = \frac{4}{2}[1, 1]$.

(c) $\mathbf{u} = [2, 1]$: $\mathbf{P}_{\mathbf{u}} = \frac{1}{5}\begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}$. For $\mathbf{v} = [1, 3]$, $\mathbf{P}_{\mathbf{u}}\mathbf{v} = \frac{1}{5}[10, 5] = [2, 1]$, and the leftover $\mathbf{v} - \mathbf{P}_{\mathbf{u}}\mathbf{v} = [-1, 2]$ is orthogonal to $\mathbf{u}$. This is the picture below.

(d) In $\mathbb{R}^3$, $\mathbf{u} = [1, 1, 1]$: $\mathbf{P}_{\mathbf{u}} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$, and $\mathbf{P}_{\mathbf{u}}\mathbf{v}$ replaces every entry of $\mathbf{v}$ by the average of the entries: $\mathbf{P}_{\mathbf{u}}[1, 2, 6] = [3, 3, 3]$.



26.4. The structure of $\mathbf{P_u}$

Read off the definition:

- **Symmetric.** $(\mathbf{uu}^\mathsf{T})^\mathsf{T} = (\mathbf{u}^\mathsf{T})^\mathsf{T}\mathbf{u}^\mathsf{T} = \mathbf{uu}^\mathsf{T}$ by M6.
- **Column space the line.** Column j of $\mathbf{uu}^\mathsf{T}$ is $u_j\mathbf{u}$, a multiple of $\mathbf{u}$. So $\mathrm{col}[\mathbf{P_u}] = \mathrm{span}\{\mathbf{u}\}$ and $\operatorname{rank}\mathbf{P_u} = 1$. In $\mathbb{R}^2$ that means $\det\mathbf{P_u} = 0$ (§14.2); e.g. $\frac{1}{5}(4\cdot 1 - 2\cdot 2) = 0$.
- **The line is kept.** $\mathbf{P_u u} = \mathbf{u}\,\frac{\mathbf{u}^\mathsf{T}\mathbf{u}}{\mathbf{u}^\mathsf{T}\mathbf{u}} = \mathbf{u}$, so every vector already on the line stays where it is.
- **The perpendicular is crushed.** If $\mathbf{w} \perp \mathbf{u}$ then $\mathbf{P_u w} = \mathbf{u}\,\frac{\mathbf{u}^\mathsf{T}\mathbf{w}}{\mathbf{u}^\mathsf{T}\mathbf{u}} = \mathbf{0}$.

In $\mathbb{R}^2$, these last two say everything. $\mathbf{P_u}$ keeps one line, crushes the perpendicular line, and since every vector is a sum of a piece on each (§29), that fixes what it does to every vector.

27. Idempotence

27.1. Doing it twice

Once $\mathbf{v}$ has been projected onto the line, it *is* on the line, and projecting it again does nothing: the shadow of a shadow is the shadow itself. An operation with this property, that doing it twice gives the same result as doing it once, is called **idempotent** (Latin *idem*, "the same"). Everyday examples:

- pressing an elevator call button that is already lit;
- sorting a list that is already sorted;
- taking an absolute value: $\big||x|\big| = |x|$;
- rounding to the nearest integer.

Each of them loses information the first time. That is what makes it idempotent: once the information is gone, there is nothing left to lose.

27.2. The definition

For a matrix, "doing it twice" is $\mathbf{PP} = \mathbf{P}^2$ (§15.4). For the projection onto a line it can be verified directly. Using M1, and that $\mathbf{u}^\mathsf{T}\mathbf{u}$ is a number,

$$\mathbf{P_u^2} = \frac{\mathbf{uu}^\mathsf{T}\,\mathbf{uu}^\mathsf{T}}{(\mathbf{u}^\mathsf{T}\mathbf{u})^2} = \frac{\mathbf{u}\,(\mathbf{u}^\mathsf{T}\mathbf{u})\,\mathbf{u}^\mathsf{T}}{(\mathbf{u}^\mathsf{T}\mathbf{u})^2} = \frac{\mathbf{uu}^\mathsf{T}}{\mathbf{u}^\mathsf{T}\mathbf{u}} = \mathbf{P_u}.$$

Definition 27.1 (Projection matrix)**.** A square matrix $\mathbf{P}$ is a **projection matrix** if it is idempotent:

$$\mathbf{P}^2 = \mathbf{P}.$$

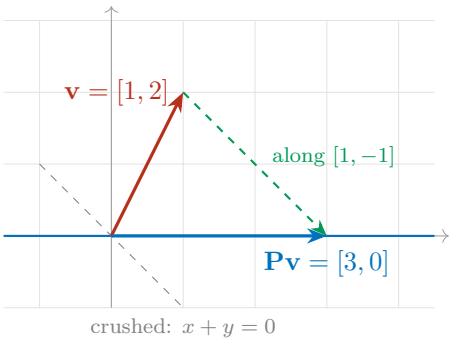
Examples: $\mathbf{I}$ and $\mathbf{0}$ (the projections that keep everything and keep nothing), $\mathrm{diag}(1, 0)$, and every $\mathbf{P_u}$. Non-examples: $\mathbf{J}$, since $\mathbf{J}^2 = -\mathbf{I}$; and $2\mathbf{I}$, since $(2\mathbf{I})^2 = 4\mathbf{I}$.

27.3. A slanted shadow

The definition says nothing about perpendicular, and it allows more than $\mathbf{P_u}$. Take

$$\mathbf{P} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \qquad \mathbf{P}^2 = \begin{bmatrix} 1+0 & 1+0 \\ 0 & 0 \end{bmatrix} = \mathbf{P}.$$

It is a projection. It sends $[\,x, y\,]$ to $[\,x + y,\ 0\,]$, so it lands every point on the x-axis and keeps points already there. But the vectors it crushes satisfy $x + y = 0$: the line through $[\,1, -1\,]$, which is not perpendicular to the x-axis. Each point slides along the direction $[\,1, -1\,]$ until it hits the axis. That is a shadow cast by light coming in at an angle.



27.4. Orthogonal projections

What separates $\mathbf{P_u}$ from the slanted example is symmetry: $\mathbf{P_u}^{\mathsf{T}} = \mathbf{P_u}$, while $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ is not symmetric.

Definition 27.2 (Orthogonal projection matrix)**.** A square matrix $\mathbf{P}$ is an **orthogonal projection matrix** if $\mathbf{P}^2 = \mathbf{P}$ and $\mathbf{P}^{\mathsf{T}} = \mathbf{P}$.

Theorem 27.1. *If* $\mathbf{P}$ *is an orthogonal projection matrix, then for every* $\mathbf{v}$ *the leftover* $\mathbf{v} - \mathbf{Pv}$ *is orthogonal to every vector of* col[$\mathbf{P}$].

Proof. A vector of col[$\mathbf{P}$] has the form $\mathbf{Pw}$. Then

$$(\mathbf{Pw})^{\mathsf{T}}(\mathbf{v} - \mathbf{Pv}) = \mathbf{w}^{\mathsf{T}}\mathbf{P}^{\mathsf{T}}\mathbf{v} - \mathbf{w}^{\mathsf{T}}\mathbf{P}^{\mathsf{T}}\mathbf{Pv} = \mathbf{w}^{\mathsf{T}}\mathbf{Pv} - \mathbf{w}^{\mathsf{T}}\mathbf{P}^2\mathbf{v} = \mathbf{w}^{\mathsf{T}}\mathbf{Pv} - \mathbf{w}^{\mathsf{T}}\mathbf{Pv} = 0,$$

using $\mathbf{P}^{\mathsf{T}} = \mathbf{P}$ and then $\mathbf{P}^2 = \mathbf{P}$. □

So for an orthogonal projection the light falls perpendicular, which is the picture of §26.3 and the reason for the name. This theorem returns in Part 9, where the best approximate solution of an unsolvable system turns out to be an orthogonal projection.

28. The complementary projection and reflections

28.1. $\mathbf{I} - \mathbf{P}$ is a projection too

If $\mathbf{P}$ is a projection, so is $\mathbf{I} - \mathbf{P}$. By the distributive laws, with $\mathbf{IP} = \mathbf{PI} = \mathbf{P}$,

$$(\mathbf{I} - \mathbf{P})^2 = \mathbf{I} - 2\mathbf{P} + \mathbf{P}^2 = \mathbf{I} - 2\mathbf{P} + \mathbf{P} = \mathbf{I} - \mathbf{P}.$$

If $\mathbf{P}$ is symmetric, so is $\mathbf{I} - \mathbf{P}$, so the complement of an orthogonal projection is an orthogonal projection. The matrix $\mathbf{I} - \mathbf{P}$ is the **complementary projection**.

28.2. What it does

For the projection onto a line,

$$(\mathbf{I} - \mathbf{P_u})\mathbf{v} = \mathbf{v} - \mathrm{proj}_{\mathbf{u}}(\mathbf{v}),$$

the orthogonal piece of the vector unit's §12.6: the part of $\mathbf{v}$ that is perpendicular to $\mathbf{u}$. The two projections split $\mathbf{v}$ between them:

$$\mathbf{P_u}\mathbf{v} + (\mathbf{I} - \mathbf{P_u})\mathbf{v} = \mathbf{v}, \qquad \mathbf{P_u}(\mathbf{I} - \mathbf{P_u}) = \mathbf{P_u} - \mathbf{P_u}^2 = \mathbf{0}, \qquad (\mathbf{I} - \mathbf{P_u})\mathbf{P_u} = \mathbf{0}.$$

The pieces add back to $\mathbf{v}$, and each projection kills the other's piece.

28.3. In the plane: the projection onto the perpendicular line

In $\mathbb{R}^2$ the perpendicular piece lies on a single line: the line through a vector $\mathbf{v} \perp \mathbf{u}$. Projecting onto that line is the same construction as §26 with $\mathbf{v}$ in the role of $\mathbf{u}$: the projection of any vector $\mathbf{x}$ onto $\mathbf{v}$ is $\mathrm{proj}_{\mathbf{v}}(\mathbf{x}) = \mathbf{P_v}\mathbf{x}$, where

$$\mathbf{P_v} = \frac{\mathbf{v}\mathbf{v}^\mathsf{T}}{\mathbf{v}^\mathsf{T}\mathbf{v}},$$

the **orthogonal projection matrix with respect to $\mathbf{v}$**. The claim is that this *is* the complement of $\mathbf{P_u}$.

Theorem 28.1. *Let $\mathbf{u} = [\,a, b\,] \neq \mathbf{0}$ and $\mathbf{v} = [\,-b, a\,]$, which is orthogonal to $\mathbf{u}$ and has the same length. Then*

$$\mathbf{P_u} + \mathbf{P_v} = \mathbf{I}, \qquad \textit{that is,} \qquad \mathbf{I} - \mathbf{P_u} = \mathbf{P_v}.$$

Proof. Add the two outer products:

$$\mathbf{u}\mathbf{u}^\mathsf{T} + \mathbf{v}\mathbf{v}^\mathsf{T} = \begin{bmatrix} a^2 & ab \\ ab & b^2 \end{bmatrix} + \begin{bmatrix} b^2 & -ab \\ -ab & a^2 \end{bmatrix} = (a^2 + b^2)\,\mathbf{I}.$$

Both have denominator $\mathbf{u}^\mathsf{T}\mathbf{u} = \mathbf{v}^\mathsf{T}\mathbf{v} = a^2 + b^2$, so dividing gives $\mathbf{P_u} + \mathbf{P_v} = \mathbf{I}$. □

Example 28.1. With $\mathbf{u} = [\,1, 1\,]$ and $\mathbf{v} = [\,-1, 1\,]$,

$$\mathbf{I} - \mathbf{P_u} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \frac{\mathbf{v}\mathbf{v}^\mathsf{T}}{\mathbf{v}^\mathsf{T}\mathbf{v}} = \mathbf{P_v}. \quad \checkmark$$

Remark. In $\mathbb{R}^3$ the complement of a line projection is not a line projection: what is perpendicular to $\mathbf{u}$ is a whole plane, and $\mathbf{I} - \mathbf{P_u}$ has rank 2. With $\mathbf{u} = [\,1, 1, 1\,]$, $\mathbf{P_u}$ replaces each entry by the average (Example 26.1(d)), so $\mathbf{I} - \mathbf{P_u}$ subtracts the average from each entry: $[\,1, 2, 6\,] \mapsto [\,-2, -1, 3\,]$. Data scientists call this *centering*, and it is a projection onto the plane of vectors whose entries sum to zero.

28.4. Reflection, from projection

A reflection across the line span$\{\mathbf{u}\}$ keeps the piece of $\mathbf{v}$ along $\mathbf{u}$ and reverses the perpendicular piece:

$$\mathbf{v} \;\longmapsto\; \mathbf{P_u}\mathbf{v} - (\mathbf{I} - \mathbf{P_u})\mathbf{v} = (2\mathbf{P_u} - \mathbf{I})\,\mathbf{v}.$$

So the reflection across span$\{\mathbf{u}\}$ is $2\mathbf{P_u} - \mathbf{I}$, which in the plane is also $\mathbf{P_u} - \mathbf{P_v}$. Check against Part 3: $\mathbf{u} = \mathbf{e}_1$ gives $2\,\mathrm{diag}(1, 0) - \mathbf{I} = \mathrm{diag}(1, -1)$, the reflection across the x-axis, and $\mathbf{u} = [\,1, 1\,]$ gives $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} - \mathbf{I} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, the reflection across $y = x$.

It is symmetric, and $(2\mathbf{P}-\mathbf{I})^2 = 4\mathbf{P}^2 - 4\mathbf{P} + \mathbf{I} = \mathbf{I}$, so it is its own transpose and squares to $\mathbf{I}$. Then $(2\mathbf{P}-\mathbf{I})^\mathsf{T}(2\mathbf{P}-\mathbf{I}) = \mathbf{I}$, and it is an orthogonal matrix (§20.6): reflecting twice puts everything back, and lengths are kept.

28.5. Reflecting a vector onto the x-axis

Here is a job for a reflection. Given a nonzero vector $\mathbf{u} \in \mathbb{R}^2$, find a reflection that sends $\mathbf{u}$ onto the positive x-axis.

Where u must land. A reflection keeps lengths (§28.4), so the image of $\mathbf{u}$ has length $\|\mathbf{u}\|$. The only point of the positive x-axis at that distance from the origin is

$$\|\mathbf{u}\|\,\mathbf{e}_1 = [\,\|\mathbf{u}\|,\,0\,].$$

Where the mirror must be. A mirror line sits halfway between a point and its reflection. So the midpoint of $\mathbf{u}$ and its image, which is the *average* (or centroid) of the two vectors,

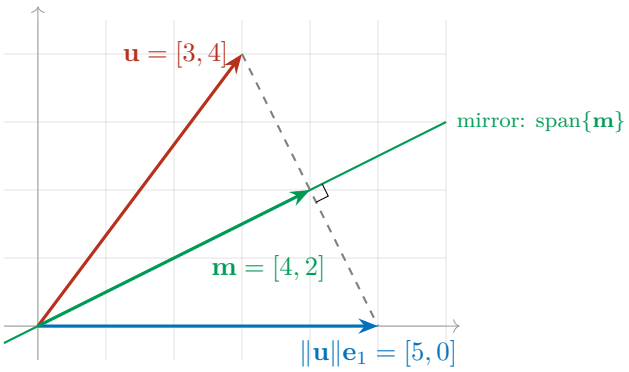
$$\mathbf{m} = \frac{\mathbf{u} + \|\mathbf{u}\|\,\mathbf{e}_1}{2},$$

lies on the mirror. The mirror is a line through the origin, so it is $\operatorname{span}\{\mathbf{m}\}$, as long as $\mathbf{m} \neq \mathbf{0}$ (§28.7 handles the one exception). There is a second way to see it. The vectors $\mathbf{u}$ and $\|\mathbf{u}\|\mathbf{e}_1$ have the same length, so the parallelogram they span is a rhombus. Its diagonal $\mathbf{u} + \|\mathbf{u}\|\mathbf{e}_1$ cuts the angle between them in half, so the mirror is the line that bisects that angle.

A check. The segment from $\mathbf{u}$ to its image should cross the mirror at a right angle. It does, because the two vectors have the same length:

$$(\mathbf{u} - \|\mathbf{u}\|\mathbf{e}_1)\cdot(\mathbf{u} + \|\mathbf{u}\|\mathbf{e}_1) = \mathbf{u}\cdot\mathbf{u} - \|\mathbf{u}\|^2\,\mathbf{e}_1\cdot\mathbf{e}_1 = \|\mathbf{u}\|^2 - \|\mathbf{u}\|^2 = 0.$$

The picture is for $\mathbf{u} = [\,3,4\,]$, with $\|\mathbf{u}\| = 5$ and $\mathbf{m} = \frac{1}{2}\big([\,3,4\,] + [\,5,0\,]\big) = [\,4,2\,]$.



28.6. The Householder reflector

Now build the reflection across $\operatorname{span}\{\mathbf{m}\}$ with §28.4. Scale $\mathbf{m}$ to the unit vector $\mathbf{v} = \mathbf{m}/\|\mathbf{m}\|$. Then $\mathbf{P}_\mathbf{m} = \mathbf{v}\mathbf{v}^\mathsf{T}$ (§26.2), and the reflection is $2\mathbf{P}_\mathbf{m} - \mathbf{I}$.

Definition 28.1 (Householder reflector)**.** Let $\mathbf{u} \in \mathbb{R}^2$ be a nonzero vector that does not point along the negative x-axis. Let

$$\mathbf{m} = \frac{\mathbf{u} + \|\mathbf{u}\|\,\mathbf{e}_1}{2} \qquad \text{and} \qquad \mathbf{v} = \frac{\mathbf{m}}{\|\mathbf{m}\|}.$$

The **Householder reflector** for $\mathbf{u}$ is

$$\boxed{\mathbf{H} = 2\mathbf{v}\mathbf{v}^{\mathsf{T}} - \mathbf{I}_2},$$

the reflection across the line spanned by the average of $\mathbf{u}$ and $\|\mathbf{u}\|\mathbf{e}_1$. Here $\mathbf{I}_2$ is the 2×2 identity.

Theorem 28.2. *The Householder reflector sends* $\mathbf{u}$ *onto the positive* x*-axis:* $\mathbf{H}\mathbf{u} = \|\mathbf{u}\|\,\mathbf{e}_1$*.*

Proof. The key fact is that the projection of $\mathbf{u}$ onto the line through $\mathbf{m}$ is $\mathbf{m}$ itself. Write $r = \|\mathbf{u}\|$, and $u_1 = \mathbf{e}_1 \cdot \mathbf{u}$ for the first entry of $\mathbf{u}$. Then

$$\mathbf{m} \cdot \mathbf{u} = \tfrac{1}{2}\big(\mathbf{u} \cdot \mathbf{u} + r\,\mathbf{e}_1 \cdot \mathbf{u}\big) = \tfrac{1}{2}\big(r^2 + ru_1\big),$$

$$\mathbf{m} \cdot \mathbf{m} = \tfrac{1}{4}\big(\mathbf{u} \cdot \mathbf{u} + 2r\,\mathbf{e}_1 \cdot \mathbf{u} + r^2\,\mathbf{e}_1 \cdot \mathbf{e}_1\big) = \tfrac{1}{4}\big(2r^2 + 2ru_1\big) = \tfrac{1}{2}\big(r^2 + ru_1\big).$$

The two are equal, so $\mathbf{P}_{\mathbf{m}}\mathbf{u} = \frac{\mathbf{m}\cdot\mathbf{u}}{\mathbf{m}\cdot\mathbf{m}}\,\mathbf{m} = \mathbf{m}$. Therefore

$$\mathbf{H}\mathbf{u} = 2\mathbf{P}_{\mathbf{m}}\mathbf{u} - \mathbf{u} = 2\mathbf{m} - \mathbf{u} = (\mathbf{u} + r\,\mathbf{e}_1) - \mathbf{u} = r\,\mathbf{e}_1. \qquad \square$$

In the language of §28.4, the piece of $\mathbf{u}$ along the mirror is $\mathbf{m}$, and the perpendicular piece is $\mathbf{u} - \mathbf{m} = \frac{1}{2}(\mathbf{u} - \|\mathbf{u}\|\mathbf{e}_1)$. The reflection keeps the first piece and reverses the second, which gives $\mathbf{m} - (\mathbf{u} - \mathbf{m}) = 2\mathbf{m} - \mathbf{u}$.

Example 28.2. Take $\mathbf{u} = [\,3, 4\,]$, as in the picture of §28.5. There $\mathbf{m} = [\,4, 2\,]$, so $\|\mathbf{m}\| = \sqrt{20}$ and $\mathbf{v} = \frac{1}{\sqrt{20}}[\,4, 2\,] = \frac{1}{\sqrt{5}}[\,2, 1\,]$. Then $\mathbf{v}\mathbf{v}^{\mathsf{T}} = \frac{1}{5}\begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}$, the projection onto the line through $[\,2, 1\,]$ from Example 26.1(c), and

$$\mathbf{H} = \frac{2}{5}\begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \frac{1}{5}\begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}, \qquad \mathbf{H}\mathbf{u} = \frac{1}{5}\,[\,9 + 16,\ 12 - 12\,] = [\,5, 0\,]. \quad \checkmark$$

28.7. Properties, and the usual form

It is a reflection. $\mathbf{H}$ is built as in §28.4, so it is symmetric, $\mathbf{H}^2 = \mathbf{I}$, and it is orthogonal. Its determinant is -1: in Example 28.2, $\frac{1}{25}(-9 - 16) = -1$.

Its first column is u, scaled to length 1. Apply $\mathbf{H}$ to both sides of $\mathbf{H}\mathbf{u} = \|\mathbf{u}\|\mathbf{e}_1$. Since $\mathbf{H}^2 = \mathbf{I}$, this gives $\mathbf{u} = \|\mathbf{u}\|\,\mathbf{H}\mathbf{e}_1$, so $\mathbf{H}\mathbf{e}_1 = \mathbf{u}/\|\mathbf{u}\|$. Every 2×2 orthogonal matrix with determinant -1 has the form $\begin{bmatrix} a & c \\ c & -a \end{bmatrix}$ (§21.2), so in the plane

$$\mathbf{H} = \frac{1}{\|\mathbf{u}\|}\begin{bmatrix} u_1 & u_2 \\ u_2 & -u_1 \end{bmatrix}.$$

This is a quick check on a computation. In Example 28.2 it reads $\frac{1}{5}\begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$. It is not a replacement for the construction, since it works only in the plane.

The exception. If $\mathbf{u}$ points along the negative x-axis, then $\mathbf{u} = -\|\mathbf{u}\|\mathbf{e}_1$ and the average $\mathbf{m}$ is $\mathbf{0}$, so there is no line to reflect across. No construction is needed: the reflection across the y-axis, $\mathrm{diag}(-1, 1)$, sends $\mathbf{u}$ to $\|\mathbf{u}\|\mathbf{e}_1$.

The usual form. Software, and most books, write a Householder reflector as $\mathbf{I} - 2\mathbf{w}\mathbf{w}^\mathsf{T}$ with $\mathbf{w}$ a unit vector. In the plane this is the same matrix. Take $\mathbf{w}$ to be a unit vector perpendicular to $\mathbf{v}$. Theorem 28.1 gives $\mathbf{v}\mathbf{v}^\mathsf{T} + \mathbf{w}\mathbf{w}^\mathsf{T} = \mathbf{I}$, so

$$2\mathbf{v}\mathbf{v}^\mathsf{T} - \mathbf{I} = 2(\mathbf{I} - \mathbf{w}\mathbf{w}^\mathsf{T}) - \mathbf{I} = \mathbf{I} - 2\mathbf{w}\mathbf{w}^\mathsf{T}.$$

By the check in §28.5, $\mathbf{w}$ points along $\mathbf{u} - \|\mathbf{u}\|\mathbf{e}_1$, the segment from the image to $\mathbf{u}$. So $\mathbf{I} - 2\mathbf{w}\mathbf{w}^\mathsf{T}$ reads "reflect across the line perpendicular to $\mathbf{w}$". That description carries over to $\mathbb{R}^n$: there the vectors perpendicular to $\mathbf{w}$ form an $(n-1)$-dimensional subspace (a plane, in $\mathbb{R}^3$), and $\mathbf{I} - 2\mathbf{w}\mathbf{w}^\mathsf{T}$ reflects across it. The reflectors are named for Alston Householder, who introduced them for this purpose in 1958.

Looking ahead. Part 11 uses a sequence of Householder reflectors to compute the QR decomposition. Each one turns a column into a multiple of $\mathbf{e}_1$, which zeros out every entry below the first, and keeps lengths exactly.

29. The direct sum

29.1. Splitting a vector

Take orthogonal nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^2$. By Theorem 28.1, every $\mathbf{x} \in \mathbb{R}^2$ splits as

$$\mathbf{x} = \mathbf{P}_\mathbf{u}\mathbf{x} + \mathbf{P}_\mathbf{v}\mathbf{x}, \qquad \mathbf{P}_\mathbf{u}\mathbf{x} \in \mathrm{span}\{\mathbf{u}\}, \quad \mathbf{P}_\mathbf{v}\mathbf{x} \in \mathrm{span}\{\mathbf{v}\} :$$

a piece on each line.

Example 29.1. With $\mathbf{u} = [\,1, 1\,]$, $\mathbf{v} = [\,-1, 1\,]$ and $\mathbf{x} = [\,3, 1\,]$: $\mathbf{P}_\mathbf{u}\mathbf{x} = [\,2, 2\,]$ and $\mathbf{P}_\mathbf{v}\mathbf{x} = \frac{1}{2}[\,2, -2\,] = [\,1, -1\,]$. Their sum is $[\,3, 1\,]$.

29.2. The split is unique

More is true: there is *no other way* to write $\mathbf{x}$ as a vector on the first line plus a vector on the second. Suppose $\mathbf{x} = \mathbf{p} + \mathbf{q} = \mathbf{p}' + \mathbf{q}'$, with $\mathbf{p}, \mathbf{p}'$ on $\mathrm{span}\{\mathbf{u}\}$ and $\mathbf{q}, \mathbf{q}'$ on $\mathrm{span}\{\mathbf{v}\}$. Then

$$\mathbf{p} - \mathbf{p}' = \mathbf{q}' - \mathbf{q}$$

is a vector on both lines. Two different lines through the origin meet only at the origin, since $\mathbf{u}$ and $\mathbf{v}$ are independent (the vector unit, §21.2) and $c\mathbf{u} = d\mathbf{v}$ forces $c = d = 0$. So $\mathbf{p} = \mathbf{p}'$ and $\mathbf{q} = \mathbf{q}'$.

29.3. The definition

Definition 29.1 (Direct sum)**.** Let U and W be subspaces of a subspace V. Then V is the **direct sum** of U and W, written

$$V = U \oplus W,$$

if every vector of V can be written in exactly one way as $\mathbf{u} + \mathbf{w}$ with $\mathbf{u} \in U$ and $\mathbf{w} \in W$.

Two things are being asked: that every vector *can* be split (the subspaces together reach everything), and that the split is *unique*. The second can be replaced by a simpler test.

Theorem 29.1. *$V = U \oplus W$ if and only if every vector of V is a sum $\mathbf{u}+\mathbf{w}$ with $\mathbf{u} \in U$, $\mathbf{w} \in W$, and $U \cap W = \{\mathbf{0}\}$.*

Proof. If $U \cap W = \{\mathbf{0}\}$, the argument of §29.2 shows splittings are unique. Conversely, if some $\mathbf{z} \neq \mathbf{0}$ lay in both, then $\mathbf{0} = \mathbf{0} + \mathbf{0} = \mathbf{z} + (-\mathbf{z})$ would be two different splittings of $\mathbf{0}$. □

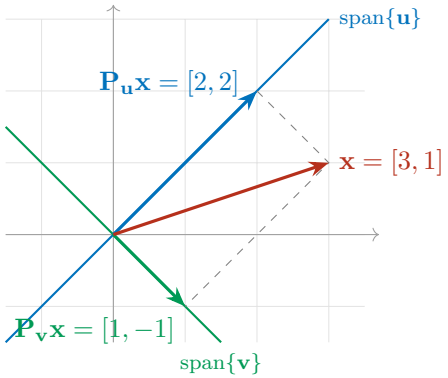
So, for orthogonal nonzero $\mathbf{u}, \mathbf{v}$,

$$\mathbb{R}^2 = \operatorname{span}\{\mathbf{u}\} \oplus \operatorname{span}\{\mathbf{v}\},$$

and the two projections $\mathbf{P_u}$ and $\mathbf{P_v} = \mathbf{I} - \mathbf{P_u}$ are the machines that find the two pieces. Because the pieces are orthogonal, their lengths obey the Pythagorean identity (the vector unit, §23):

$$\|\mathbf{x}\|^2 = \|\mathbf{P_u x}\|^2 + \|\mathbf{P_v x}\|^2.$$

In Example 29.1, $10 = 8 + 2$.



29.4. Direct sums need not be orthogonal

Nothing in the definition mentions angles. The slanted projection of §27.3, $\mathbf{P} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$, and its complement $\mathbf{I} - \mathbf{P} = \begin{bmatrix} 0 & -1 \\ 0 & 1 \end{bmatrix}$ split $\mathbf{x} = [\,x, y\,]$ as

$$\mathbf{x} = \underbrace{[\,x+y,\ 0\,]}_{\text{on the } x\text{-axis}} + \underbrace{[\,-y,\ y\,]}_{\text{on } \operatorname{span}\{[-1,1]\}},$$

so $\mathbb{R}^2 = \operatorname{span}\{[\,1, 0\,]\} \oplus \operatorname{span}\{[\,-1, 1\,]\}$, two lines at 135°. For $\mathbf{x} = [\,3, 1\,]$ the pieces are $[\,4, 0\,]$ and $[\,-1, 1\,]$. They are not orthogonal, and the Pythagorean identity fails: $10 \neq 16 + 2$. A direct sum needs only that the two subspaces reach everything and meet only at $\mathbf{0}$; orthogonality is an extra, and a valuable one.

29.5. Every projection splits the space

The pattern is general. For any projection $\mathbf{P} \in \mathbb{R}^{n \times n}$,

$$\mathbb{R}^n = \operatorname{col}[\mathbf{P}] \oplus \operatorname{col}[\mathbf{I} - \mathbf{P}].$$

Every $\mathbf{x}$ splits as $\mathbf{Px} + (\mathbf{I}-\mathbf{P})\mathbf{x}$, a vector of each. And the two meet only at $\mathbf{0}$: if $\mathbf{z} = \mathbf{Pa} = (\mathbf{I}-\mathbf{P})\mathbf{b}$, then on one hand $\mathbf{Pz} = \mathbf{P}^2\mathbf{a} = \mathbf{Pa} = \mathbf{z}$, and on the other $\mathbf{Pz} = \mathbf{P}(\mathbf{I}-\mathbf{P})\mathbf{b} = \mathbf{0}$, so $\mathbf{z} = \mathbf{0}$. Conversely, given any direct sum $\mathbb{R}^n = U \oplus W$, the rule "keep the U-piece" is linear and idempotent, so it is a projection. Projections and direct sums are two descriptions of the same thing.

Example 29.2. In $\mathbb{R}^3$, with $\mathbf{u} = [\,1,1,1\,]$, $\mathbb{R}^3 = \operatorname{span}\{\mathbf{u}\} \oplus \{\mathbf{x} : x_1 + x_2 + x_3 = 0\}$, a line and the plane perpendicular to it. The vector $[\,1,2,6\,]$ splits as $[\,3,3,3\,] + [\,-2,-1,3\,]$, its average and its centered part.

29.6. Practice

P1. Write $\mathbf{P_u}$ for $\mathbf{u} = [\,3,4\,]$, and use it to project $[\,1,0\,]$ and $[\,4,-3\,]$.

P2. Which of these are projection matrices, and which are orthogonal projection matrices?

$$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \qquad \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}, \qquad \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}, \qquad \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}.$$

P3. For $\mathbf{u} = [\,1,2\,]$, compute $\mathbf{I} - \mathbf{P_u}$ and check that it equals $\mathbf{P_v}$ with $\mathbf{v} = [\,-2,1\,]$.

P4. With the same $\mathbf{u}$ and $\mathbf{v}$, split $\mathbf{x} = [\,4,3\,]$ into a piece on each line and check the Pythagorean identity.

P5. Find the matrix of the reflection across the line through $[\,1,2\,]$, and reflect $[\,1,0\,]$.

P6. Show that if $\mathbf{P}$ is a projection then $\mathbf{P}^k = \mathbf{P}$ for every $k \geq 1$.

P7. Find the Householder reflector $\mathbf{H}$ for $\mathbf{u} = [\,-3,4\,]$. Check that $\mathbf{Hu} = [\,5,0\,]$ and that the first column of $\mathbf{H}$ is $\mathbf{u}/\|\mathbf{u}\|$. Compare with P5.

Answers

P1. $\mathbf{P_u} = \frac{1}{25}\begin{bmatrix} 9 & 12 \\ 12 & 16 \end{bmatrix}$. $[\,1,0\,] \mapsto \frac{1}{25}[\,9,12\,]$, and $[\,4,-3\,] \mapsto \mathbf{0}$, since $[\,4,-3\,] \perp [\,3,4\,]$.

P2. $\operatorname{diag}(0,1)$: idempotent and symmetric, orthogonal projection onto the y-axis. $\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$: its square is itself, so it is a projection, but it is not symmetric. It sends $[\,x,y\,]$ to $[\,x,x\,]$, onto the line $y = x$ along vertical lines, a slanted one. $\frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \mathbf{P}_{[1,1]}$: orthogonal projection. $\operatorname{diag}(2,0)$: its square is $\operatorname{diag}(4,0)$, not a projection.

P3. $\mathbf{P_u} = \frac{1}{5}\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$, so $\mathbf{I} - \mathbf{P_u} = \frac{1}{5}\begin{bmatrix} 4 & -2 \\ -2 & 1 \end{bmatrix}$, and $\mathbf{vv}^{\mathsf{T}}/\mathbf{v}^{\mathsf{T}}\mathbf{v} = \frac{1}{5}\begin{bmatrix} 4 & -2 \\ -2 & 1 \end{bmatrix}$. ✓

P4. $\mathbf{P_u x} = \frac{4+6}{5}[\,1,2\,] = [\,2,4\,]$ and $\mathbf{P_v x} = \frac{-8+3}{5}[\,-2,1\,] = [\,2,-1\,]$. Sum $[\,4,3\,]$; $25 = 20 + 5$. ✓

P5. $2\mathbf{P_u} - \mathbf{I} = \frac{2}{5}\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} - \mathbf{I} = \begin{bmatrix} -\frac{3}{5} & \frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{bmatrix}$, and $[\,1,0\,] \mapsto [\,-\frac{3}{5}, \frac{4}{5}\,]$. (A reflection of the form $\begin{bmatrix} a & c \\ c & -a \end{bmatrix}$, as §21 requires.)

P6. By induction: $\mathbf{P}^{k+1} = \mathbf{P}^k\mathbf{P} = \mathbf{PP} = \mathbf{P}$.

P7. $\|\mathbf{u}\| = 5$, so $\mathbf{m} = \frac{1}{2}\big([\,-3,4\,] + [\,5,0\,]\big) = [\,1,2\,]$ and $\mathbf{v} = \frac{1}{\sqrt{5}}[\,1,2\,]$. Then

$$\mathbf{H} = \frac{2}{5}\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} - \mathbf{I} = \begin{bmatrix} -\frac{3}{5} & \frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{bmatrix}, \qquad \mathbf{Hu} = \Big[\,\frac{9+16}{5},\ \frac{-12+12}{5}\,\Big] = [\,5,0\,].$$

The first column is $[\,-\frac{3}{5}, \frac{4}{5}\,] = \mathbf{u}/5$. ✓ It is the matrix of P5: the mirror is the line through $[\,1,2\,]$, and P5's $[\,1,0\,] \mapsto [\,-\frac{3}{5}, \frac{4}{5}\,]$ is the first-column fact, $\mathbf{He}_1 = \mathbf{u}/\|\mathbf{u}\|$.

Looking ahead. Part 6 takes the two subspaces that a projection splits the space into, col[$\mathbf{P}$] and what $\mathbf{P}$ crushes, and asks the same questions of every matrix. What does it reach? What does it crush to $\mathbf{0}$? The second set is the null space, and the rank–nullity theorem says how the two sizes are related.